Appendix for

Tossing a Coin: Preferences for Rank-ordering in Triage Decisions

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Data availability statement

The data analyses were conducted in R. The data set and our main analysis file can be accessed via our GitHub repository (https://github.com/Peer-Kessler/tossing_a_coin).

Sample

Our sample was recruited via the UK based online panel provider Prolific (www.prolific.com). Participation in our study was voluntary and all participants gave informed consent to the use of their answers for scientific purposes.

All Prolific users from North America and Europe were eligible for participation in our study. The study was shown to all eligible users until we reached $1,000$ participants from North America and $1,000$ participants from Europe. Since participants from the UK and the US are overrepresented in the Prolific panel, we limited the number of users from the UK to 100 participants and the users from the US to 800 participants. Our final sample consists of $2,283$ subjects who participated in our study. We excluded all participants who failed any of three attention check questions (12.1%). This left the failed participants' positions in the survey vacant and other users were able to participate. Additionally, we excluded all participants who finished the survey in under 5 minutes after data collection and before the analysis (10.6%). Pretesting the survey revealed that finishing the survey in under 5 minutes is almost impossible. This is partly due to the fact that participants were forced to stay on the description of the fictional situation for at least 30 seconds before they were able to continue.

After data cleaning, the final sample consisted of $1,998$ completed surveys. Table A1 shows the distribution of self-reported countries in our sample. One person reported to reside in the United Arab Emirates even though we restricted the participation to persons from North America and Europe. However, we are not interested in country-specific differences and did not exclude this person from our sample. The same holds true for one person who did not report her country of residence.

Table A1: Sample by self-reported country of residence.

Country	n	%
Austria	2	0.10
Belgium	12	0.60
Canada	202	10.11
Caribbean Netherlands	1	0.05
Czech Republic	19	0.95
Denmark	6	0.30
Finland	11	0.55
France	29	1.45
Germany	12	0.60
Greece	61	3.05
Hungary	34	1.70
Ireland	17	0.85
Italy	112	5.61
Netherlands	22	1.10
Norway	3	0.15
Poland	224	11.21
Portugal	252	12.61
Spain	82	4.10
Sweden	4	0.20
Switzerland	2	0.10
United Arab Emirates	1	0.05
United Kingdom	100	5.00
United States	787	39.39
United States Minor Outlying Islands	2	0.10
Missing	1	0.05
Sum	1998	100.00

Attention checks

To ensure high quality answers, we included three attention check questions in our study at three different positions. First, we included a question with an obvious answer stating that its purpose is to identify bots at the beginning of the survey (Dunn et al., 2018):

What shape has a red circle?

Please provide the correct answer. This is a quality question to identify bots.

☐ Rectangular

☐ Round (*Correct answer*)

☐ Green

☐ Red

Right before the conjoint experiment, the participants were presented an instructional text in which the correct answer was hidden (Oppenheimer et al., 2009). The question text was included in the survey as an image to avoid responses by bots:

Recent research on decision making shows that choices are affected by context. Differences in how people feel, their previous knowledge and experience, and their environment can affect choices. We are interested in whether you actually take the time to read the directions. To

show that you have read the instructions, please ignore the question below about how you are feeling and instead check the None of these option as your answer and continue. Thank you very much.

Please check the word that describes best how you are currently feeling.

☐ Afraid

☐ Excited

☐ Concerned

☐ None of these (*Correct answer*)

For the third attention check, participants were simply asked to select *I fully agree* on a 6-point rating scale (Berinsky et al., 2014). This question was shown after the conjoint experiment and the triage evaluation, and just before the demographic questions:

Please select "I fully agree" to indicate that you are paying attention to the questions.

☐ I do not agree at all

☐ I do not agree

☐ I rather do not agree

☐ I rather agree

☐ I agree

☐ I fully agree (*Correct answer*)

Conjoint experiment

As an introduction to the conjoint experiment, the participants were shown a fictional situation.

> *Please imagine the following situation:*
> *A pandemic has been spreading for several months. The virus is transmitted by droplets in the air. Vaccination is accessible to all; it is voluntary and protects others as well. It also reduces the risk of serious illness for the vaccinated person himself. After a steep rise in infections, the number of emergency patients with life threatening respiratory conditions is increasing. The national health care system has reached its limits. There are more patients than ventilators. Physicians must now decide who should and who should not be ventilated or kept on a ventilator. Patients who are not ventilated have effectively no chance of survival and can only be given pain relief. Imagine that you are a member of a hospital ethics committee and that you are advising the physicians in the intensive care unit on these difficult and stressful decisions. In the following, you will be presented with different situations, one after the other. In each of these situations there are two patients but only one available ventilator. The second ventilator is to be moved to another room*

in the intensive care unit where it is urgently needed.
After 30 seconds you can proceed with the survey.

To ensure that participants take the time to read the text, we included a timer at the bottom of the survey page that counts down from 30 seconds. Only after these 30 seconds was the button to continue shown. Firstly, participants were presented with five pairs of profiles from which they had to choose one of the two patients for ventilation. Secondly, they were introduced to the possibility of additionally allocating the respiratory ventilator by random chance.

You already made 5 choices. In the following you will be presented with 5 more situations. However, now you have the additional option to leave the selection to random chance.
If you leave your decision to random chance, then, as in a coin toss, each of the two patients has an equal chance of being selected for ventilation or continued ventilation.

Figure A3 shows an example of a pair of profiles with the option to allocate the ventilator by random chance. The patients' profiles consisted of eight attributes which were randomly drawn from a multivariate uniform distribution. Consequently, all variables that refer to patients' attributes are uniformly distributed and statistically independent qua experimental design (Hainmueller et al., 2014). However, the occurrence of two identical profiles in a pair of profiles was suppressed in the programming. Therefore, the profiles always differed in at least one attribute. Additionally, the order in which the attributes were presented was randomized between participants and was held constant for the whole conjoint experiment. An overview of all attributes and the corresponding attribute levels in our conjoint experiment can be found in Table A2.

Please choose which person should be ventilated or leave the selection to random chance.
Please review the information on both patients thoroughly and make a recommendation.

	Person A	**Person B**
Age	75 years old	25 years old
Weight	Not overweight	Overweight
Volunteer work	Does volunteer work	Does not volunteer work
Vaccination status	Is vaccinated	Is vaccinated
Children	Has children	Has children
Chance of survival with ventilation	80% chance of survival	20% chance of survival
Disability	Has a disability	Has a disability
Ventilation status	Is already ventilated	Is already ventilated

○ Person A ○ Person B ○ Random Chance

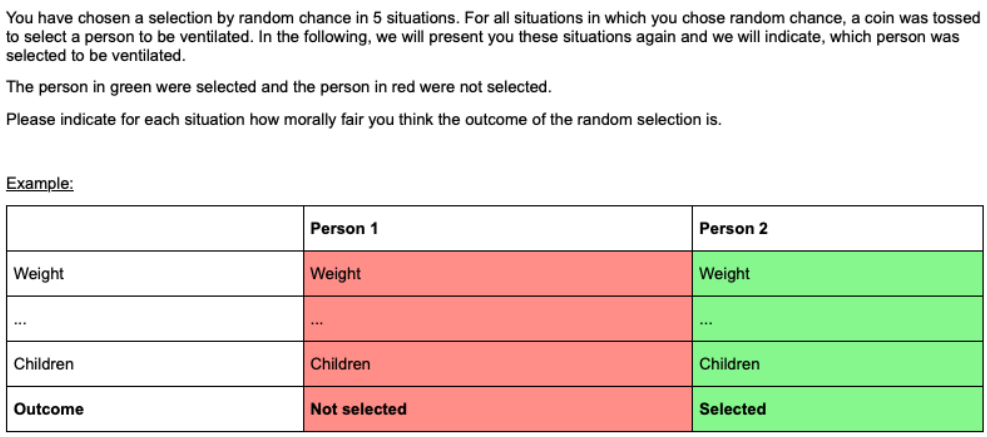
Fig. A1: Example of a pair of profiles with the option to allocate the ventilator by random chance.

Table A2: Attributes and Attribute Levels in conjoint experiment.

Attribute	Attribute Level
Chance of survival with ventilation	20% chance of survival 50% chance of survival 80% chance of survival
Age	25 years old 50 years old 75 years old
Disability	Has no disability Has a disability
Overweight	Not overweight Overweight
Ventilation status	Is not yet ventilated Is already ventilated
Vaccination status	Is not vaccinated Is vaccinated
Volunteer work	Does not volunteer work Does volunteer work
Children	Has no children Has children

Fairness ratings

Every pair of profiles in which the participants chose the allocation by random chance was shown to the participants again. This time we indicated the outcome of the random lottery by color-coding the selected profiles. Selected profiles are displayed in green, and unselected profiles are displayed in red. The participants were shown the instructions and an example beforehand (see Figure A2). The participants were asked to rate the perceived fairness of each outcome on a 6-point rating scale. Figure A3 shows an example.



You have chosen a selection by random chance in 5 situations. For all situations in which you chose random chance, a coin was tossed to select a person to be ventilated. In the following, we will present you these situations again and we will indicate, which person was selected to be ventilated.

The person in green were selected and the person in red were not selected.

Please indicate for each situation how morally fair you think the outcome of the random selection is.

Example:

	Person 1	**Person 2**
Weight	Weight	Weight
...	...	...
Children	Children	Children
Outcome	**Not selected**	**Selected**

Fig. A2: Introduction and example given to the participants before the fairness rating of outcomes by random chance.

Please indicate how morally fair you think the outcome of the random selection is.

	Person A	Person B
Weight	Not overweight	Overweight
Disability	Has a disability	Has a disability
Ventilation status	Not ventilated yet	Is already ventilated
Age	75 years old	50 years old
Children	Has children	Has children
Chance of survival with ventilation	20% chance of survival	20% chance of survival
Vaccination status	Is vaccinated	Is vaccinated
Volunteer work	Does not volunteer work	Does volunteer work

I find the outcome of the random selection...

not fair at all. very fair.

Fig. A3: Example of a pair of profiles with rating question.

Since all profiles were randomly generated, we used a fixed selection of profiles to indicate the outcome of a selection by random chance. If a participant choose the random-chance option for the first of the five pairs of profiles, profile B was subsequently shown as selected for the fairness rating. For the second and third pair of profiles, profile A was selected. For the fourth pair of profiles, profile B was indicated as selected, and for the last pair of profiles, profile A was indicated as selected. Please recall that pairs of profiles were only shown for the fairness rating if the participants chose a selection by random chance for the pair of profiles.

Robustness check

Average marginal mean estimates

Additional to AMCEs on deliberate decisions for one of the patients (see Figure 2 in the main text), we estimated average marginal means (Leeper et al., 2020). As can be seen in Figure A4, the overall interpretation if results remains the same. Most importantly, patients that have lower chances of survival or are older in age are by and large the most disadvantaged patients independent from the other attributes. By contrast, patients with higher chances of survival and that are younger are chosen more likely by the participants.

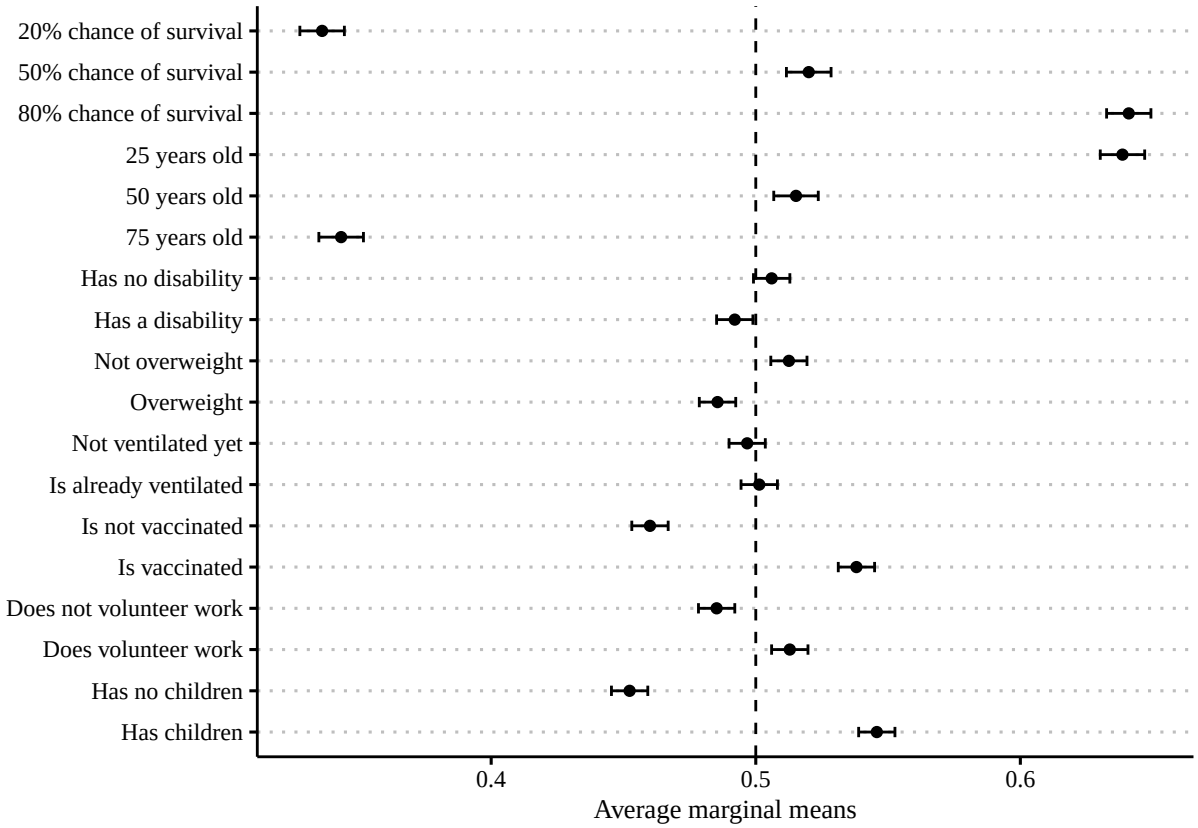


Fig. A4: Average marginal means estimates for linear model on the deliberate choice between two patients; bars represent 95% confidence intervals; the model includes patient-level random effects (N = 1,905); the model is based on 15,106 statistical units.

Model specification

Since the fairness ratings of the outcomes of a selection by random chance are measured on an ordinal scale, we calculated ordered logistic regression as a robustness check. More specifically, we used a cumulative link mixed model (CLMM) to integrate normally distributed participant-level random effects to account for the hierarchical structure of our data (Christensen, 2019). Figure A5 shows the results from the linear OLS regression with clustered standard errors on the participant level and the results from the CLMM. We reported the logit coefficients for the CLMM for better comparability.

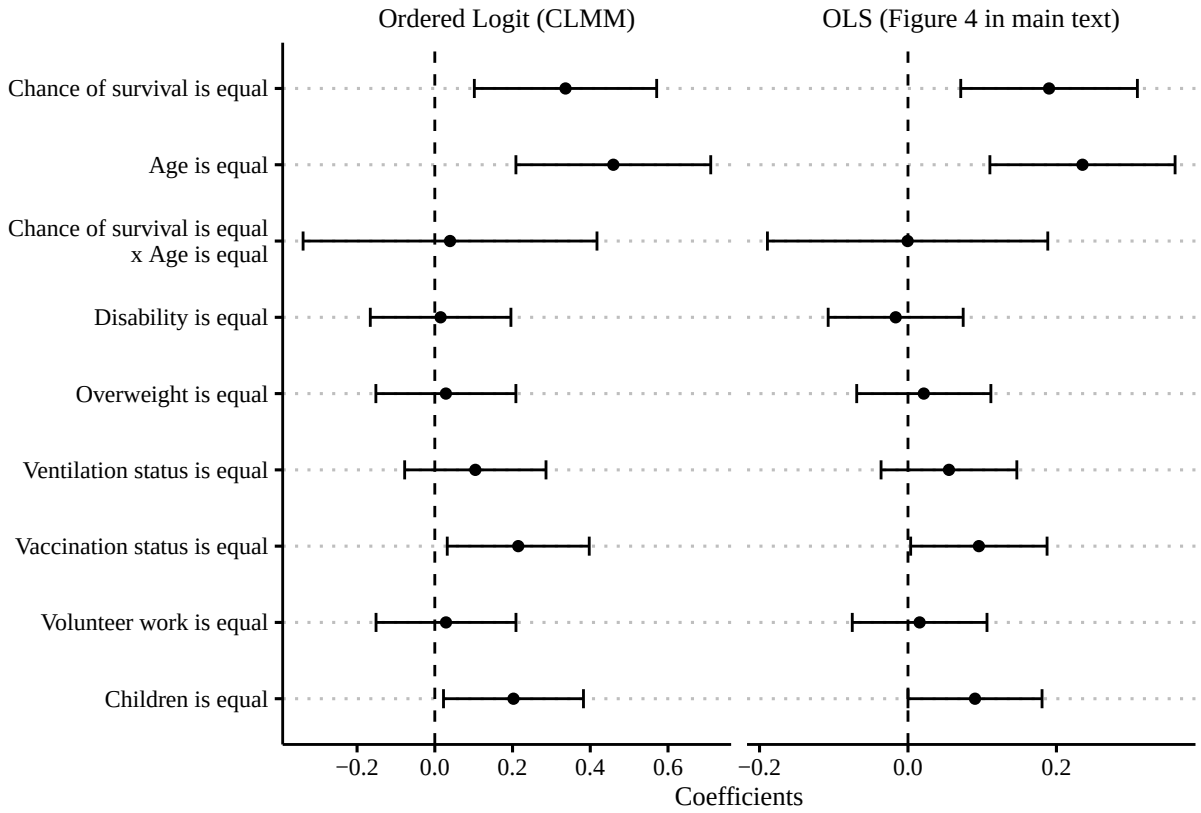


Fig. A5: Robustness check for fairness rating; bars represent 95% confidence intervals; both models include patient-level random effects (N = 1,231); both models are based on 2,437 statistical units.

Regression tables

Table A3: Regression table for Figure 2 in the main article.

	Dependent variable:
	Choice of a profile (dummy)
	Linear OLS
Chance of survival (Ref.: 20% chance of survival)	
50% chance of survival	0.182***
	(0.009)
80% chance of survival	0.312***
	(0.009)
Age (Ref.: 25 years old)	
50 years old	−0.134***
	(0.009)
75 years old	−0.312***
	(0.009)
Disability (Ref.: Has no disability)	
Has a disability	−0.016**
	(0.008)
Overweight (Ref.: Is not overweight)	
Is overweight	−0.022***
	(0.008)
Ventilation status (Ref.: Not ventilated yet)	
Is already ventilated	0.009
	(0.008)
Vaccination status (Ref.: Is not vaccinated)	
Is vaccinated	0.079***
	(0.008)
Volunteer work (Ref.: Does not volunteer work)	
Does volunteer work	0.014*
	(0.008)
Children (Ref.: Has no children)	
Has children	0.092***
	(0.008)
Constant	0.403***
	(0.012)
Observations	15,106
Log Likelihood	−9,775.409
Akaike Inf. Crit.	19,576.820
Bayesian Inf. Crit.	19,675.920
Note:	*p<0.1; **p<0.05; ***p<0.01

Table A4: Regression table for Figure 3 and Figure 4 in the main article, and Figure A5 in this Appendix.

	Figure 3	Figure 3	Figure A5
	Dependent variable:		
	Choice of random-lottery option	Fairness rating	Fairness rating
	Logit (GLMM)	*Linear OLS*	*Ordered logit (CLMM)*
Chance of survival is equal	0.571***	0.190***	0.336***
	(0.075)	(0.060)	(0.120)
Age is equal	0.170**	0.235***	0.459***
	(0.077)	(0.064)	(0.128)
Chance of survival is equal x Age is equal	0.801***	−0.001	0.039
	(0.127)	(0.096)	(0.193)
Disability is equal	−0.015	−0.017	0.015
	(0.058)	(0.046)	(0.092)
Overweight is equal	0.023	0.021	0.029
	(0.058)	(0.046)	(0.092)
Ventilation status is equal	0.177***	0.055	0.104
	(0.058)	(0.047)	(0.093)
Vaccination status is equal	0.348***	0.095**	0.215**
	(0.059)	(0.047)	(0.093)
Volunteer work is equal	0.003	0.016	0.029
	(0.058)	(0.046)	(0.092)
Children is equal	0.186***	0.090*	0.202**
	(0.058)	(0.046)	(0.092)
Constant	-2.272***	3.883***	
	(0.098)	(0.073)	
1|2			-3.898
			(0.199)
2|3			-2.543
			(0.168)
3|4			-1.078
			(0.153)
4|5			1.198
			(0.152)
5|6			2.937
			(0.170)
Observations	9,990	2,437	2,437
Log Likelihood	−5029.273	−-3,899.661	−3653.627
Akaike Inf. Crit.	10,080.545	7,823.323	7,337.255
Bayesian Inf. Crit.	10,159.848	7,892.905	7,424.233
Note:			*p<0.1; **p<0.05; ***p<0.01

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